

Druid: (Re)connecting with Nature through Computational Fashion Wearables

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Figure 1: Final Design of Druid: A. User's virtual avatar in a virtual forest. B. Minimal sock C. Flamboyant gaiter.

Abstract

Nature Connectedness (NC) is directly linked to an individual's well-being and a sense of an Ecological Self that includes the natural environment and feelings of care towards it. However, modern day life has largely severed our connection with nature, removing an integral part of our being, a source of meaning, and our awareness of how our actions are affecting the climate. As a potential solution, we present Druid, a Computational Fashion Wearable (CFW) designed to help foster NC and redevelop the bond between people and nature. Druid is a wearable ensemble of vibrating socks and fashionable gaiters linked to a virtual avatar and a virtual forest environment in Virtual Reality (VR). The design safely guides the user to points of interest in real forests, encouraging observation and active engagement with their surroundings. This in turn impacts the virtual forests, calling for further nature exploration. This

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design has been evaluated in terms of function, visual design and comfort, and potential future directions. Results positively indicate that Druid could already be applied in practice, and the broader concept of fostering NC through the use of CFWs and VR has many positive potential applications.

CCS Concepts

• **Human-centered computing** → **Interaction devices; Interaction design process and methods.**

Keywords

Nature, Virtual Reality, Wearables, Computational Fashion Wearable, Research through Design

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1 Introduction

Engagement with nature has long been associated with well-being, offering various physiological and psychological benefits [25], [3]. Directly linked to well-being, Nature Connectedness (NC) increases

one's happiness and can provide a sense of meaning in life [14, 17]. Further, it develops the Ecological Self that includes the natural environment and feelings of care and love towards it [4]. However, in today's increasingly technology-driven world, humans are becoming disconnected from nature, leading to a growing "nature deficit" that impacts both individual and planetary health [23]. The observed disconnect is particularly problematic concerning climate change mitigation, as fomenting a sense of ecological responsibility requires immersive contact with nature [13].

While technology might be seen as a contributor to this disconnect, it may be a possible remedy. Many mobile applications have been developed to encourage users to go outside [12], or to provide them with more knowledge about their surroundings [21]. However, screen-based interactions may work against the immediacy and embodied experience offered by the actual contact with nature, taking attention away from it [13]. Virtual experiences have been used to foster nature and climate change awareness [8, 26]. However, these experiences too often lack the depth and sensory richness of physical interactions. Active engagement is deemed as critical for fostering NC [19], underlining the critical need for research in interfaces that can smoothly bridge physical existence of forests with digital interfaces and thus provide a way to interact with an environment in more meaningful and embodied ways.

We see Computational Fashion Wearables (CFWs) as a potential solution that could extend technological solutions used in nature, and connect them to virtual experiences, taking advantage of their unique affordances. CFWs have the ability to facilitate new ways of interacting with the environment through tactile and audiovisually communicative interfaces [10], and utilize the body in novel and directed ways to promote sensory interactions [7]. By being tangible extensions of the user, they can ground virtual experiences in the physical world without sacrificing the unique affordances of self-expression that VR offers. Wearables, with unique position in fashionability and self-expression can also reinforce identity congruence, ensuring that the virtual and physical selves are aligned [6, 16, 20] with potential to further transform wearer's identities for developing kinship with nature and the Ecological Self.

Here, we present a CFW design - Druid: a wearable ensemble of vibrating socks and fashionable gaiters linked to a virtual avatar and a virtual forest environment in VR. The design safely guides the user in nature to points of interest, encouraging observation and active engagement with their surroundings, and provides further insight and reflection on their experiences by re-framing them in VR, calling for further nature exploration. Our findings through the preliminary evaluations of the design (in terms of function, visual design and comfort, and overall concept) indicate that Druid could already be applied in practice, and the broader concept of fostering NC through the use of CFWs and VR has many potential future applications. Our findings suggest that playful wearables can be an excellent direction for facilitating physical presence and embodied interaction with forests. Equally, a virtual forest connected to it can provide pleasurable and educational experiences for increasing knowledge, awareness and connectedness to nature.

2 Background and Related Work

2.1 Nature Connectedness, Technological Approaches and Limitations

Numerous studies indicate the benefits of connecting with natural landscapes, inducing positive effects on mental and physical health [25], stress-reduction, and mood and cognitive improvements [2, 3]. Furthermore, developing a relationship with the natural world creates ecological awareness, resulting in sustainable behavior in activities like creating food forests and conservation activities [18]. In this regard, Nature Connectedness, a psychological construct explaining an individual's sensory and affective relationship with nature, is considered important for sustainability [19]. However, modern ways of life have been accompanied by increasing distance from nature, producing a relationship in which people often perceive nature as something strange or to be dominated, rather than as part of themselves. This "distance" feeds exploitation, ignorance, and, finally, climate change, because people lack the knowledge and emotional bond that would provide the motivation to digest and act on information about the environment [13].

Technological advances have been viewed as one of the possible avenues for reconnecting humans with nature. Mobile apps, for example, facilitate such interaction by providing guided tours in nature, species identification, and environmental learning, which enhances users' knowledge and appreciation of their surroundings [1, 11, 15]. Similarly, virtual environments (e.g., video games and VR), expose people to nature in very experiential ways, offering experiences such as virtual forests or ecosystems that educate and inspire users about biodiversity and conservation [5]. Such approaches can foster environmental awareness and engagement, especially given limited access to natural spaces.

However, current methods often rely on screens, which can limit embodiment and fail to achieve a deep, sensory connection with nature [13]. Screen-mediated interactions lack the tactility and immersion of physical experiences, thus rendering them less effective for developing meaningful connections with the environment. This calls for the development of more embodied and interactive methodologies, closing the chasm between the physical and the virtual nature and allowing users to engage in their surrounding environment through hands-free, immersive experiences.

2.2 Computational Fashion Wearables: The Bridge Between Virtual and Physical

Human-nature interaction through clothing and fashion is a deeply rooted concept, as garments have traditionally served as mediators between individuals and their environment. For instance, traditional clothing often represents cultural relationships with nature, such as the use of natural materials or designs inspired by local ecosystems [9]. CFWs extend this concept by incorporating environmental sensors and smart materials that make it possible for wearables to interact with and augment natural environments. For example, a gaiter with sensors integrated can guide a person through a forest, thus enriching the experience by providing real-time feedback on environmental conditions such as temperature, humidity, and even the existence of wild life. Not only does this bring the physical and digital worlds closer together, but it also brings the person closer to

nature, enabling them to develop environmental awareness and a sense of curiosity. Through the integration of fashion, technology, and nature, CFWs present a distinctive opportunity to reconceptualize the interactions between humans and their environments. These systems allow individuals to interact with their surroundings in significant manners, whether by utilizing augmented reality enhancements within a forest or by converting natural occurrences into wearable responses. This synthesis of fashion and technology serves to not only elevate personal expression but also fosters a more balanced relationship among humans, technology, and the natural environment [24]. In this respect, as CFWs evolve, they might redefine what we consider and interact with in both the physical and digital landscapes, opening up possibilities for a future where fashion functions not only as expression but also as a bridge for deeper ecological and technological engagement. Druid unites human and nature by fusing fashion, technology, and nature engagement. It deepens sensory experience and environmental awareness, demonstrating the ways in which CFWs have the ability to transform human-nature interactions in the age of technology.

3 Design Process

Our team is composed of individuals with expertise in fashion, design, VR and CFWs development, and 3D character and environment art creation. Additionally, we all commonly share an affinity for nature and observe a lack of active engagement with it in our daily lives, all having ideas for different projects that could address this issue. Given our shared interests, our design process focused on addressing the *Good Health and Well-being* and *Climate Action* UN Sustainable Development Goals. We approached these by aiming to increase individuals' affective and sensory connection to nature (NC) for fostering users' well-being and motivating sustainability actions[19]. To achieve this, we used a bottom-up research-through-design (RtD) approach [28] to explore opportunities around CFWs and VR.

We started exploring the design space, aiming to uncover how CFWs might provide continuity for both self-expression and sensations across physical and virtual spaces. We carried out brainstorming sessions using whiteboards, maintaining a detailed record of our ideation and design process on Miro board¹ which allowed us to capture every stage of the design process, from initial concept creation to the final design. **Exploring embodied and locative feedback through CFW:** Early in the process, we focused on utilizing the strength of CFWs to provide various types of feedback (e.g., auditory, visual, tactile). We considered a *design strategy of fostering curiosity through ambiguous feedback on CFWs to invite exploration and engagement with forests*. This was to augment experiences in nature by creating a sense of mystery, where users would feel there is something more to their surroundings, motivating embodied explorations in the forests. For this, we identified gloves, a necklace, a walking cane, and shoe inlays as potential choices for the CFW. For example, tactile feedback gloves could create a sensation of users touching a virtual object they cannot see, or a necklace could produce mysterious sounds to feed users' curiosity. Another strategy we explored was to *use ambiguous feedback to*

guide users to points of interest in forests to complete a task. Here, we landed on a concept for a sock and gaiter CFWs combination that would guide users in nature to points of interest where they would have to complete a task. We felt that task completion would provide educational value, whereas a foot-worn wearable would leave the hands free for interactions. For instance, guidance would be based on vibrations in a sock, and users would receive feedback through LEDs installed in the gaiter for task completion. For the tasks, we wanted users to engage with nature and encourage observation (e.g., through scanning plants or completing identification quizzes).

Exploring VR for motivating learning and nature engagements: In addition to fostering curiosity and task completion in local forest engagements, we explored how VR environments could be utilised for motivation and learning. Here, we explored a few strategies: One of them was to *use mysterious feedback in forest engagements to lead users VR to learn more about that location*: For example, traversing certain locations in the forest could trigger audible whispers, and the users would learn that these whispers belonged to a will-o-wisp when they visited the virtual forest in VR. This dichotomous approach would ensure that the CFW not only motivates the user at the time of the activity but elevates it into a meaningful, introspective narrative in VR. Connected to task completion, we considered the strategy of *using elements in VR as rewards for nature engagement, a place for reflection and learning more about the forest experiences*. For instance, for each completed task, users would unlock an experience in VR, where they can engage with a virtual forest while embodying an avatar. Task completion would trigger features of the avatar or the environment itself to change, facilitating continuous interaction and keep user's curiosity about what their virtual avatar and forest could become if they continue playing. In addition, the CFWs worn in VR would provide the same sensations, and the shoe guard would be styled to visually resonate with the avatar to provide a clear connection between the virtual and physical worlds.

4 Druid: Final Design

The final design of Druid (Fig 1) includes an ensemble of vibrating socks and fashionable gaiters linked to a virtual avatar and a virtual forest environment in VR. It incorporates the design strategies (see design process) to encourage engagement with local forests: when the user wears it while hiking in the local forest, the CFW guides the user with ambiguous feedback to points of interest by vibrating in the direction of the target location (Fig. 1 A). This non-exact guidance aims to foster exploration, engagement with forest. When they reach, the user are encouraged to actively engage with and observe their surroundings. For example, guided to discover a non-native plant, the LEDs turn on, warning the wearer to scan and correctly identify it by observing its features (Fig. 1 B). As users complete more tasks, more virtual elements and activities become available in VR (e.g., new areas, ability to plant new seeds in the VR forest, exchanging those with friends, or choosing new avatar accessories), motivating users to go back to the forest.

The modular design of the sock and the gaiter allows for the sock to be worn both in and outside of VR experiences, enabling the use of the same vibration guidance mechanic in both realities, providing

¹Available at: https://miro.com/app/board/uXjVLMgtBCY=?share_link_id=471710385444

continuity between them. Continuity is further reinforced by similar activities of searching for and discovering points of interest. The stylization of the box also provides aesthetic continuity, retaining a visual element of the gaiter while it is not worn. Similar design of gaiter and avatar provides continuity in expression between virtual and physical, and further roots the wearer with their immediate natural surroundings. VR environment and interaction takes full advantage of VR affordances to present the forest interactively not as a digital twin, but artistically reinterpreted, inspiring wonder and awe. Further, it encourages the user to reflect on what they have observed in the physical forest to notice how these elements have changed in VR. Additionally, this reinterpretation allows to represent information that may otherwise be difficult to intuitively grasp. For example, rediscovering a non-native plant in VR allows showing how non-native species may become invasive and overtake the environment, revealed to the user in just minutes.

4.1 Implementation

We identified the key moving parts of the design and the overall system. This included the physical items (sock and gaiter) with electronics for its operation, the virtual assets (avatar and environment), and an online service component accessible via a mobile app to perform tasks in nature and mediate data exchange. We prioritized creating the physical and virtual assets first, predicting the mobile app and rest of the digital infrastructure would be out of scope.

The virtual avatar and the gaiter are inspired by an interpretation of Sampsä Pellervoinen [27], a Finnish folklore figure associated with vegetation and plants, symbolising the essence of growth and renewal. In this re-imagined version, Sampsä is depicted covered in leaves and flowers. Thus, the gaiter's 3D-printed silver leaves are inspired by rowan tree leaves (in Finnish mythology symbolizing the mother of all plants on Earth [22]) reminiscent of Finnish Winters with blooming elements. The virtual avatar has their limbs similarly covered with leaves and flowers. Equally, the environment is also stylized to look more whimsical and magical as opposed to a photorealistic rendition of a forest.

To guide the user in nature, we chose to use off-the-shelf components, including vibration modules connected to DRV2605L haptic driver chips, and a LIS3MDL direction sensor to sense the magnetic North. These components were connected to a Seeed Studio XIAO nRF52840 microcontroller. We used NeoPixel LEDs in the gaiter, which would also connect to the microcontroller. We decided to offload geo-location to a mobile app that would connect to the microcontroller via Bluetooth BLE and exchange location data. In future, production-ready iterations of the design, the app could also serve the function of transferring data about physical experiences to VR by connecting to an online service. However, this was ruled to be out of scope for the current implementation. We designed the electronics housing as a silver leaf-shaped box, to which the vibrations modules and gaiter LED strip would connect to.

4.2 Team Contribution

The first author, a fashion designer with experience in wearable technology, led the design process and brought fashion principles into the process while envisioning the gaiter as a tool for augmented nature interaction. Their experience in creating wearables

that improve the connection between humans and the environment directed the design, combining style and functionality to lead the wearer through the forest. The second author, with a strong background in technology and VR development, guaranteed technical feasibility, enabling seamless interaction between the physical wearables and the virtual environment. The third author, with an expertise in avatar and virtual fashion, brought the physical and digital worlds together, creating a cohesive aesthetic for the avatar and gaiter. The second and third authors both contributed to writing the manuscript, each articulating different aspects of the technical and creative sides of the design. The fourth author designed the immersive VR forest environment. The fifth author provided supervision, assisted in meaning-making, and reviewed the process to ensure coherence and quality.

5 Initial Evaluations

We conducted two initial evaluation studies: The goal of the study 1 was to test how effectively users could distinguish different directions using our guidance mechanism and to get feedback on the CFW's design and comfort, whereas the second study focused on a semi-structured interview with forestry experts about our design concept.

5.1 Study 1: Guidance via Vibrations, Design and Comfort

The first study was conducted with 4 users (2 men and 2 women, all in their 30s) who were selected through an invitation within our research group. All participants are researchers by profession and share a common interest in visiting forests as a hobby. The experimental setup consisted of 4 vibration modules to the participant's ankle equidistantly representing cardinal directions (Fig. 2 A) and 16 directions to be signified - cardinal, intercardinal, and secondary intercardinal, represented as angles in degrees starting from the front and running clockwise. For each cardinal direction, the corresponding module was set to vibrate at 100% strength, whereas the others were deactivated. For all other target directions dir_t , the two closest modules were set to vibrate. Both modules had their output strength multiplied by a weight $w = |dir_t - dir_c|/90$, where dir_c represents the cardinal direction of the corresponding module.

For each study, users were seated and vibration modules were attached to their ankles. First, to familiarize users with the vibration sensations, users felt the directions signified individually (one second activation, one second delay) starting with the front of the ankle, moving clockwise, then returning counter-clockwise, then repeating this pattern one more time. Users were then informed they will feel individual directions being activated (one second activation, one second delay, repeated 3 times), after which they were to guess the direction before the next trial. 32 trials were carried out per user, signifying each direction twice. We then asked four users to wear the physical artifacts and provide their input on design and comfort.

5.1.1 Results about Guidance via Vibration. Results indicate users may confuse directions that are next to each other, but may find it easier to localize the direction by aligning any of the cardinal vibration modules with dir_t . We conclude that given 4 vibration modules, more than 4 distinct directions can be signified effectively.

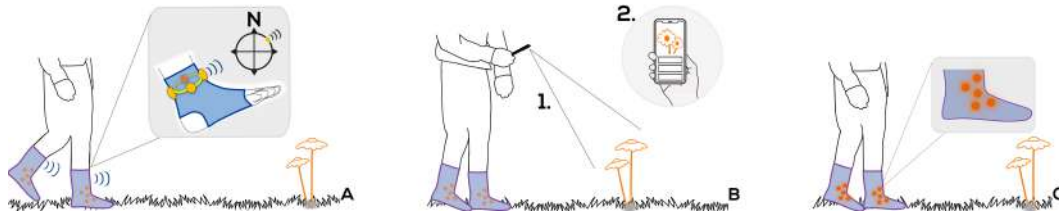


Figure 2: Nature Scenario: A. User in nature guided by vibrations around their ankle, looking for a plant B. User finds the plant, scanning it (1.) and playing an identification game based on their observations (2.) C. The gaiter lights up signifying task completion

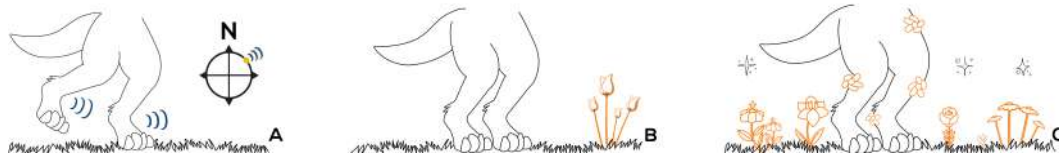


Figure 3: VR Scenario: A. User in VR guided in a virtual forest, embodying their avatar B. User finds the virtual plant, unlocked by completing a task in nature C. More similar plants begin blooming around the avatar, and on their body

Table 1: Mean angle errors in degrees for all 16 tested directions

Directions	Lowest Recorded	Mean of All Participants	Largest Recorded
All	19.34	21.62	26.02
Cardinal	11.25	15.12	22.5
Intercardinal	16.88	25.66	42.19
Secondary Intercardinal	17.58	22.85	25.31

5.1.2 Results about Design and Comfort. Most found the shoe guard comfortable. The blue-gray-white color scheme was praised for its winter aesthetic, though one of the participants recommended autumn colors for seasonal adaptability. The design was compared to trendy moon boots, which added to its stylistic appeal. The participants also appreciated the gender-neutral design, which made it versatile for a wide variety of users. The sound produced by movement, mimicking the rustling of leaves, was highlighted as an immersive addition that made one feel like walking through the forest. However, a silent mode was also suggested for it to be more versatile. Concerns over the durability of the leaf-like elements were raised, which could break if stepped on. Suggestions included improving the front design to ensure it touches the ground and providing options for customizable lengths or shapes. Another interest was the creation of a more sustainable shoe guard, possibly through the use of recycled materials (e.g., melted plastic bottles). These observations have shown the importance of balancing functionality, aesthetics, and sustainability in future versions.

5.2 Study 2: Expert Input

We conducted 4 semi-structured interviews with experts working in Forestry with industry and research experience. They were introduced to the general concept of our design and its actualized implementation - Druid. Following this, they were asked to provide their thoughts on both the design and the general concept: How could they be applied in practice, extended in the future, and what benefits they could bring. We grouped their input into 4 overarching themes.

5.2.1 Gamification. Interviewees responded positively to a gamified "collection" game approach. Druid was compared to existing games like Pokemon Go or the practice of Geo-caching. In Geo-caching, it was noted that users tend to craft additional rules and objectives to be completed, extending the basic premise of the game - such community-based evolution of an initial concept could favorably be applied to our solution. Additionally, giving people a mechanic to log qualitative data (e.g., leave a message after task completion), could quickly lead to using it as a creative outlet, with users sharing their feelings and what meaning they have assigned to their nature experiences.

5.2.2 Data Collection. Interviewees highlighted the opportunity of crowdsourcing data collection through Druid: By performing tasks, users could collect data to estimate biomass and habitat, and create maps of species distribution. The design could be applied to specific locations, such as for interacting with a national park. Park rangers could set where the users should go based on the data they need collecting, and also use the guidance mechanism as a way to ensure user safety, keeping them on the trail. Equally, users may also share useful information with one another, such as potential good foraging spots, and encouraging altruistic behavior.

5.2.3 Community and Social Interaction. Experts noted that the guiding mechanism alone could already be useful to, for example, forest surveyors working in the field, or by trail hikers and foragers to navigate the terrain safely. However, self-expression opportunities via virtual avatars and design of the wearables may also interest other communities (e.g., cosplayers and furies) that may often not be associated with forests, or engage with the aforementioned forest-goers that often. Druid could become a common point of interest, as these communities could participate in the same collection game, and meet in social VR spaces. Knowing what communities use the local forests is useful for decision-making, as parties that use the forest spaces could be consulted and represented (e.g., when creating laws and regulations).

5.2.4 Nature Augmentation. People may assign meaning to their experiences, locations and even individual trees. The affordances of CFWs to make their interaction more tangible and the ability to further reveal them in VR may strengthen the connections individuals form with their surroundings. Additionally, Extended Reality headsets may be used to see magical elements in the forest without delaying their reveal. However, some concerns were raised about technology affordability, as specialized equipment may be expensive and not very common. However, this may be less of a concern in cases where this solution is applied to specific areas (e.g., national parks) and thus equipment is provided to users.

6 Conclusion, Limitations, and Future Work

This paper introduced a novel design concept for CFWs, Druid, that aims to (re) connect us with nature by bridging virtual and physical realities. It is designed to make user experiences - virtual or physical- more ecologically minded, intertwining local mythologies to stir curiosity and provide a tool for bridging physical and virtual interactions. Combining fashion, technology, and storytelling, the Druid builds a transfiguring experience that physically bonds humanity and nature. This is the kind of relationship that can help overcome global problems such as climate change and thus would also affect personal well-being. This relates to United Nations Sustainable Development Goals number 3, health and well-being, along with number 13, which has its primary emphasis in climate action.

However, our work represents only an initial exploration of this design space. The prototypes were developed using off-the-shelf components with basic signal processing, limiting the granularity of data capture. Future iterations should leverage advanced sensors and custom software to enhance functionality. Similarly, the constraints of existing VR platforms restricted the scope of virtual design; bespoke VR applications could overcome these barriers. Further, the preliminary evaluations in this paper should only be considered as assessing our approach's potentials for meeting its aims and for getting feedback for further development and ideation, rather than any formal assessment of performance or user attitudes. Future research should work on refining these concepts for these use-cases to ensure they are not only technologically advanced but also examine in what ways they might further foster Nature Connectedness [19].

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